



ELECTRO MAGNETI C FI ELD

THEORY (EMFT)

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ELECTRO MAGNETI C

THEORY

Energy i n Mat e r i a l s

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ELECTRO MAGNETIC THEORY Materials

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ELECTRO MAGNETIC THEORY

Materials discussed



The different materials are discussed in this session namely:

- Conductors
- Resistance
- Dielectric
- Capacitance

ELECTRO MAGNETIC THEORY

Conductors



Electron
drift
velocity

$$u_e = -\mu_e E$$

hole drift velocity $u_h = \mu_h E$

Conducting current

$$J = J_e + J_h = \rho_{ve} u_e + \rho_{vh} u_h = (-\rho_{ve} \mu_e + \rho_{vh} \mu_h) E,$$

$$\sigma = -\rho_{ve} \mu_e + \rho_{vh} \mu_h,$$

Point form of Ohm's law

$$J = \sigma E,$$

ELECTRO MAGNETIC THEORY

Conductors Contd.,



A wire of diameter 1 mm and conductivity 5×10^7 S/m has 10^{29} free electrons/m³ when an electric field of 10 mV/m is applied. Determine

- (a) The charge density of free electrons
- (b) The current density
- (c) The current in the wire
- (d) The drift velocity of the electrons. Take the electronic charge as $e = -1.6 \times 10^{-19}$ C.

ELECTRO MAGNETIC THEORY

Conductors and Semiconductors



(In this particular problem, convection and conduction currents are the same.)

$$(a) \rho_v = ne = (10^{29})(-1.6 \times 10^{-19}) = -1.6 \times 10^{10} \text{ C/m}^3$$

$$(b) J = \sigma E = (5 \times 10^7)(10 \times 10^{-3}) = 500 \text{ kA/m}^2$$

$$(c) I = JS = (5 \times 10^5) \left(\frac{\pi d^2}{4} \right) = \frac{5\pi}{4} \cdot 10^{-6} \cdot 10^5 = 0.393 \text{ A}$$

$$(d) \text{ Since } J = \rho_v u, u = \frac{J}{\rho_v} = \frac{5 \times 10^5}{1.6 \times 10^{10}} = 3.125 \times 10^{-5} \text{ m/s.}$$

ELECTRO MAGNETIC THEORY

Assignment



The free charge density in copper is $1.81 \times 10^{10} \text{ C/m}^3$. For a current density of $8 \times 10^6 \text{ A/m}^2$, find the electric field intensity and the drift velocity.

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THEORY

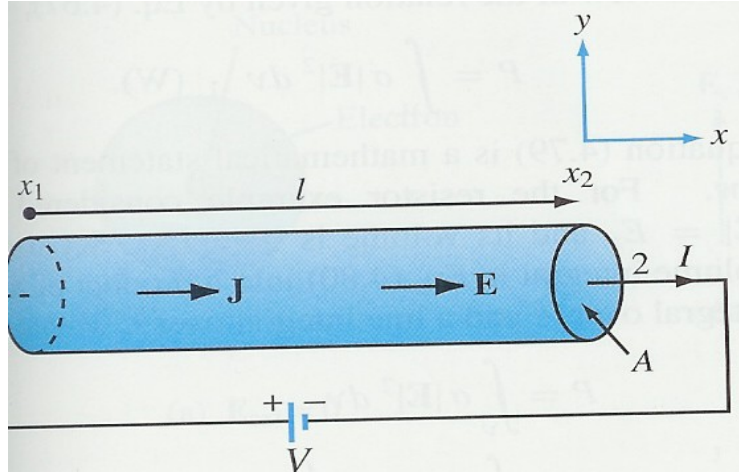
Pol ar i zat i on, El ect r i c Suscept i bi l i t y

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ELECTRO MAGNETIC THEORY

Resistance



$$I = \int_A \mathbf{J} \cdot d\mathbf{s} = \int_A \sigma \mathbf{E} \cdot d\mathbf{s} = \sigma E_x A,$$

$$R = \frac{V}{I} = \frac{- \int_{x_2}^{x_1} \mathbf{E} \cdot d\mathbf{l}}{\int_A \mathbf{J} \cdot d\mathbf{s}} = \frac{- \int_{x_2}^{x_1} \mathbf{E} \cdot d\mathbf{l}}{\int_A \sigma \mathbf{E} \cdot d\mathbf{s}},$$

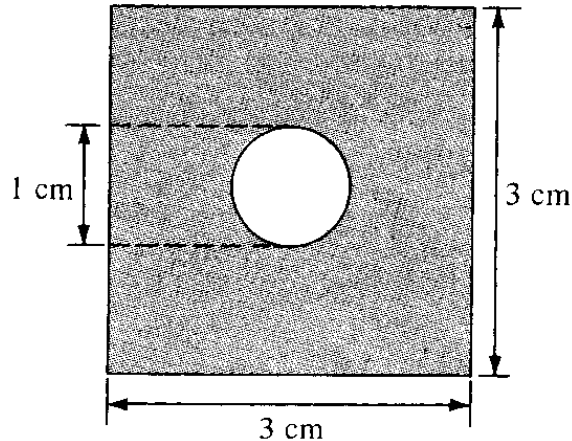
$$R = \frac{V}{I} = \frac{l}{\sigma A},$$

$$V = V_1 - V_2 = - \int_{x_2}^{x_1} \mathbf{E} \cdot d\mathbf{l} = E_x l,$$

ELECTRO MAGNETIC THEORY

Resistance Contd. ,

A lead ($\sigma = 5 \times 10^6 \text{ S/m}$) bar of square cross section has a hole bored along its length of 4 m so that its cross section becomes that of Figure shown below. Find the Resistance of the material.



$$R = \frac{\ell}{\sigma S}$$

where $S = d^2 - \pi r^2 = 3^2 - \pi \left(\frac{1}{2}\right)^2 = 9 - \frac{\pi}{4} \text{ cm}^2$.

Hence,

$$R = \frac{4}{5 \times 10^6 (9 - \pi/4) \times 10^{-4}} = 974 \mu\Omega$$

If the hole in the lead bar of Example 5.4 is completely filled with copper ($\sigma = 5.8 \times 10^6 \text{ mhos/m}$), determine the resistance of the composite bar.

ELECTRO MAGNETIC THEORY

Joule's Law



$$\Delta W = F_e \Delta l_e + F_h \Delta l_h,$$

$$\Delta P = \frac{\Delta W}{\Delta t} = F_e \frac{\Delta l_e}{\Delta t} + F_h \frac{\Delta l_h}{\Delta t} \rightarrow$$

$$= \vec{F}_e \cdot \vec{u}_e + \vec{F}_h \cdot \vec{u}_h = (\rho_{ve} \vec{u}_e E + \rho_{vh} \vec{u}_h E) \Delta v$$

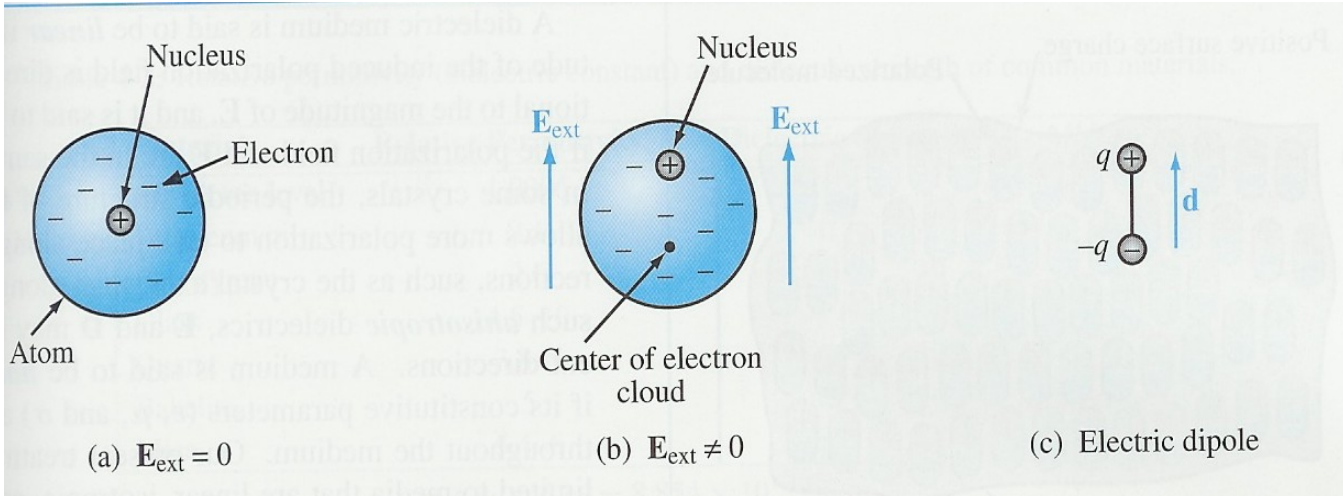
$$= \vec{J} \cdot \vec{E} \Delta v,$$

General form

$$P = \int \vec{J} \cdot \vec{E} dv,$$

ELECTRO MAGNETIC THEORY

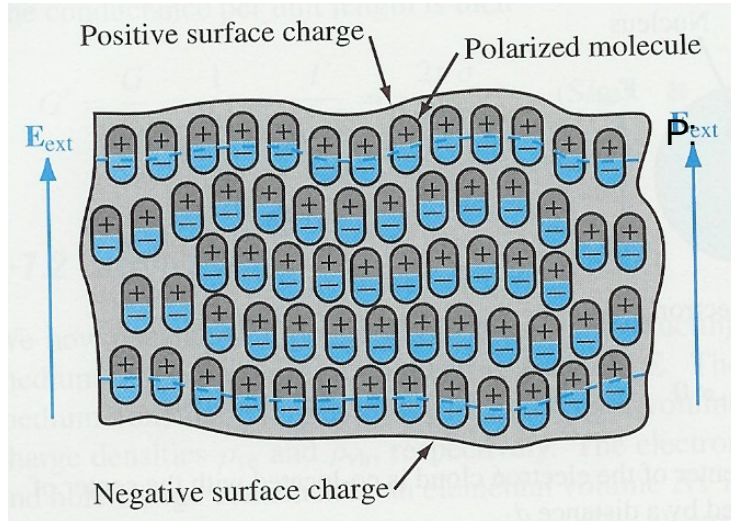
Dielectrics and Polarization



Electrical field induced
polarization

ELECTRO MAGNETIC THEORY

Electric Susceptibility



$$D = \epsilon_0 E + P,$$

electric polarization field

For homogeneous material

$$P = \epsilon_0 \chi_e E,$$

$$D = \epsilon_0 E + P = \epsilon_0 E + \epsilon_0 \chi_e E = \epsilon E,$$

$$\epsilon = \epsilon_0 (1 + \chi_e),$$

electric susceptibility

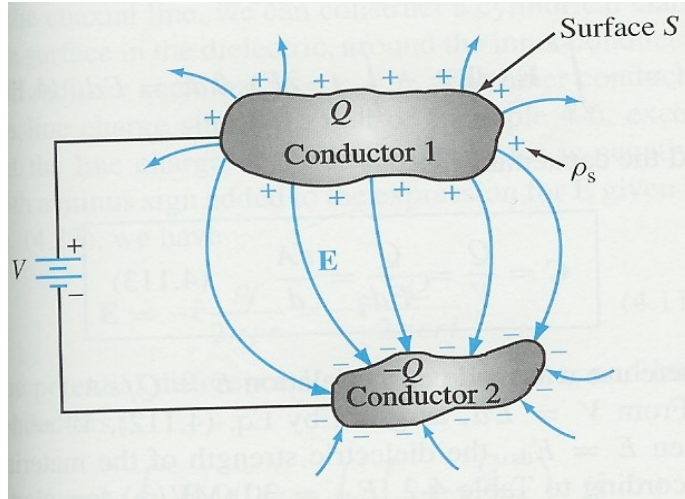
Electrical field induced

polarization

relative permittivity $\epsilon_r = \frac{\epsilon}{\epsilon_0} = (1 + \chi_e),$

ELECTRO MAGNETIC THEORY

Capacitance



$$C = \frac{Q}{V},$$

$$Q = \int \epsilon E \cdot ds,$$

$$V = - \int E \cdot dl$$

$$C = \frac{\int \epsilon \vec{E} \cdot d\vec{s}}{- \int E \cdot dl},$$

$$R = \frac{V}{I} = \frac{- \int_{x_2}^{x_1} E \cdot dl}{\int_A J \cdot ds} = \frac{- \int_{x_2}^{x_1} E \cdot dl}{\int_A \sigma E \cdot ds},$$

$$RC = \frac{\epsilon}{\sigma},$$

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Numerical Examples

The electric field intensity in polystyrene ($\epsilon_r = 2.55$) filling the space between the plates of a parallel-plate capacitor is 10 kV/m. The distance between the plates is 1.5 mm. Calculate:

- (a) D
- (b) P
- (c) The surface charge density of free charge on the plates
- (d) The surface density of polarization charge
- (e) The potential difference between the plates

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Numerical Examples Contd. ,

$$(a) \ D = \epsilon_0 \epsilon_r E = \frac{10^{-9}}{36\pi} \cdot (2.55) \cdot 10^4 = 225.4 \text{ nC/m}^2$$

$$(b) \ P = \chi_e \epsilon_0 E = (1.55) \cdot \frac{10^{-9}}{36\pi} \cdot 10^4 = 137 \text{ nC/m}^2$$

$$(c) \ \rho_s = \mathbf{D} \cdot \mathbf{a}_n = D_n = 225.4 \text{ nC/m}^2$$

$$(d) \ \rho_{ps} = \mathbf{P} \cdot \mathbf{a}_n = P_n = 137 \text{ nC/m}^2$$

$$(e) \ V = Ed = 10^4 (1.5 \times 10^{-3}) = 15 \text{ V}$$

ELECTRO MAGNETIC THEORY

Assignment



A parallel-plate capacitor with plate separation of 2 mm has a 1-kV voltage applied to its plates. If the space between its plates is filled with polystyrene ($\epsilon_r = 2.55$), find E , P , and ρ_{ps} .

ELECTRO MAGNETIC THEORY

Boundary Conditions Between Dielectric and Dielectric Media

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ELECTRO MAGNETI C THEORY

Boundary Condi t i o n s

So far we have considered the existence of the electric field in a homogeneous medium

If the field exists in a region consisting of two different media, the conditions are called **Boundary Condi t i o n s**.

These conditions are helpful in determining the field on one side

ELECTRO MAGNETIC THEORY

Boundary Conditions Contd.,

We shall consider the boundary conditions at the interface separating the following media.

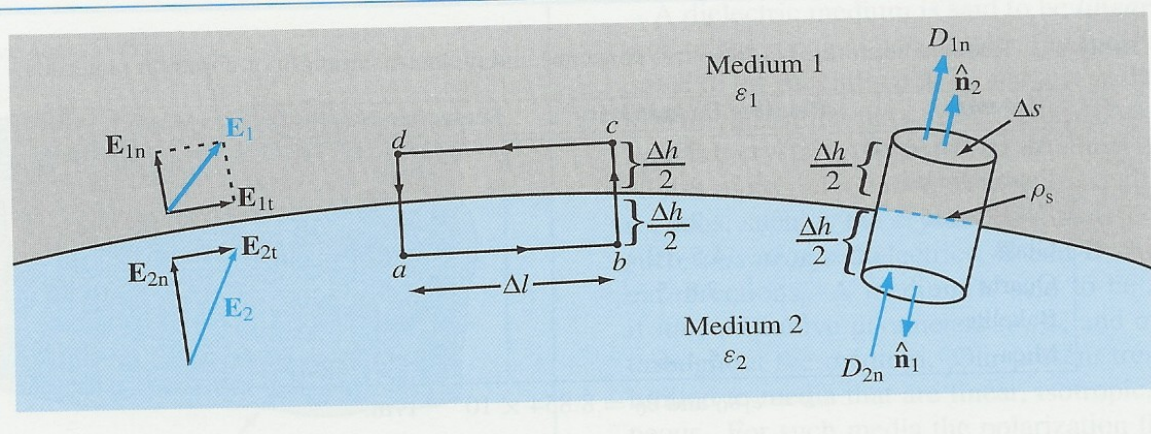
Dielectric and Dielectric

Conductor and Dielectric and

Conductor and Free Space

ELECTRO MAGNETIC THEORY

Boundary Conditions Contd.,



The tangential component is continuous across the boundary of two media.

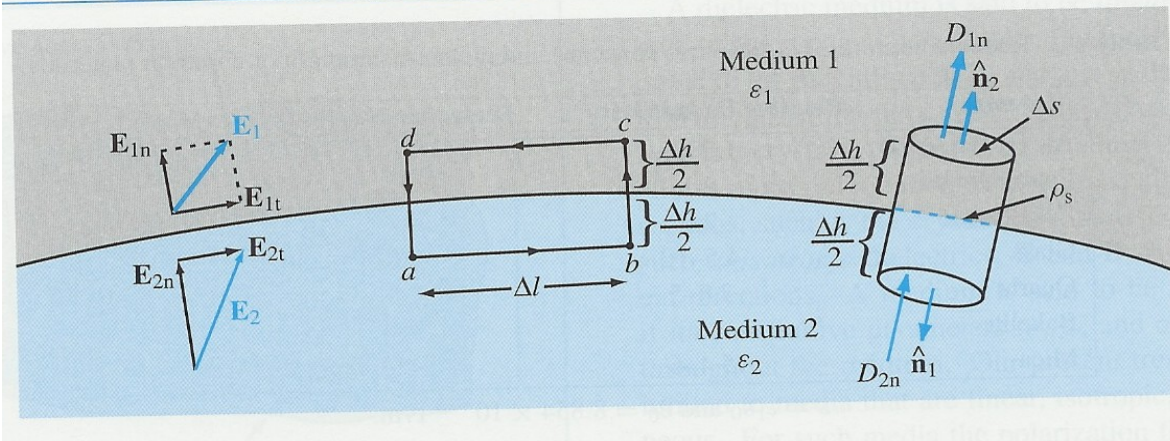
$$\oint \mathbf{E} \cdot d\mathbf{l} = \lim_{\Delta h \rightarrow 0} \left[\int \mathbf{E}_2 \cdot d\mathbf{l} + \int \mathbf{E}_1 \cdot d\mathbf{l} \right] = 0;$$

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n}, \quad \mathbf{E}_{1t} \Delta l - \mathbf{E}_{2t} \Delta l = 0,$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}, \quad \mathbf{E}_{1t} = \mathbf{E}_{2t},$$

ELECTRO MAGNETIC THEORY

Boundary Conditions



The normal component of D changes, the amount of change is equal to the surface Charge density.

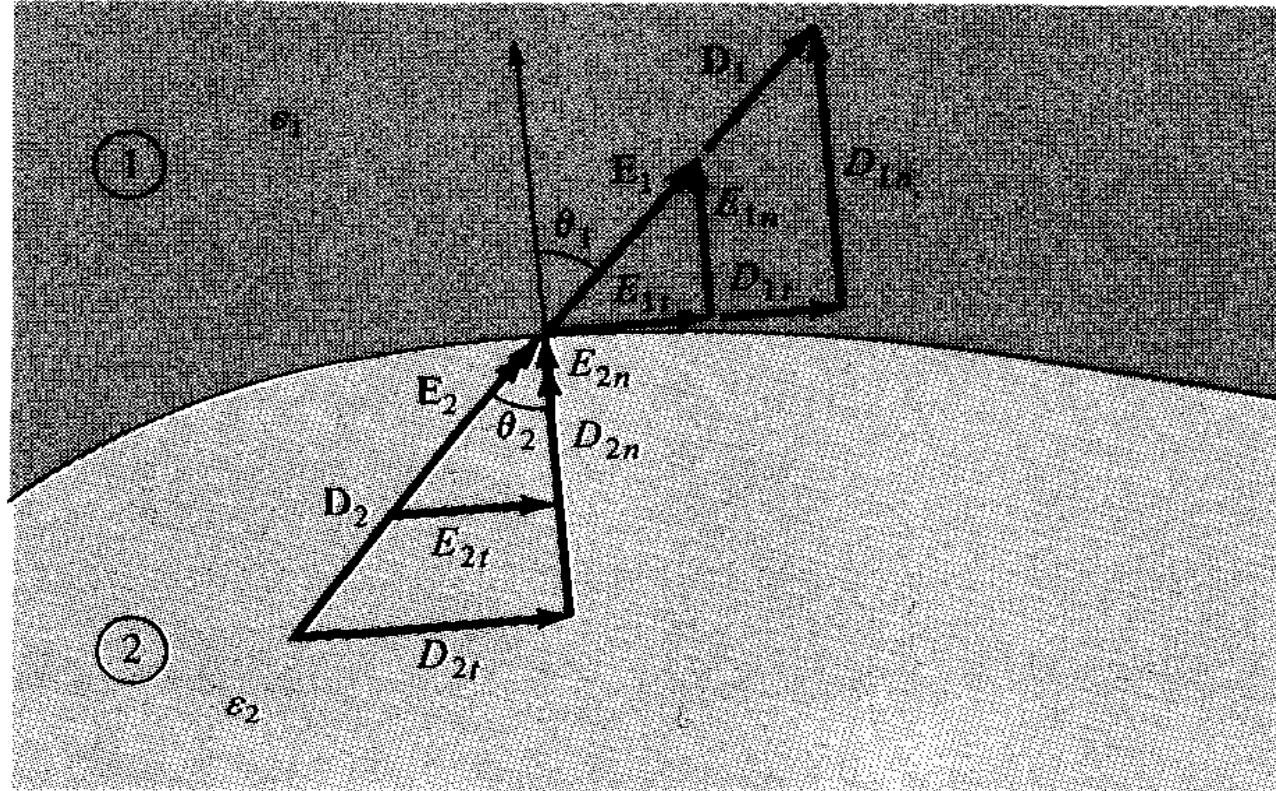
$$\oint_C \mathbf{D} \cdot d\mathbf{s} = \lim_{\Delta h \rightarrow 0} \left[\int_{top} \mathbf{D} \cdot d\mathbf{s} + \int_{bottom} \mathbf{D} \cdot d\mathbf{s} \right] = \rho_s \Delta s;$$

$$D_{1n} \Delta s - D_{2n} \Delta s = \rho_s \Delta s,$$

$$D_{1n} - D_{2n} = \rho_s,$$

ELECTRO MAGNETIC THEORY

Boundary Conditions



ELECTRO MAGNETIC THEORY

Boundary Conditions

For charge free
conditions

$$D_{1n} = D_{2n}$$

$$\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

$$E_1 \sin \theta_1 = E_{1t} = E_{2t} = E_2 \sin \theta_2$$

$$\epsilon_1 E_1 \cos \theta_1 = D_{1n} = D_{2n} = \epsilon_2 E_2 \cos \theta_2$$

$$\epsilon_1 E_1 \cos \theta_1 = \epsilon_2 E_2 \cos \theta_2$$

ELECTRO MAGNETIC THEORY

Boundary Conditions

For charge free
conditions

$$\frac{\tan \theta_1}{\epsilon_1} = \frac{\tan \theta_2}{\epsilon_2}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$$

This is the law of refraction of the electric field at a boundary free of charge (since $\rho_s = 0$ is assumed at the interface)

ELECTRO MAGNETIC THEORY

Boundary Conditions Between Dielectric, Conductor and Free Space

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ELECTRO MAGNETIC THEORY

Boundary Conditions

For Charge Free
conditions

$$\frac{\tan \theta_1}{\epsilon_1} = \frac{\tan \theta_2}{\epsilon_2}$$

$$\boxed{\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}}$$

This is the law of refraction of the electric field at a boundary free of charge (since $\rho_s = 0$ is assumed at the interface)

ELECTRO MAGNETIC THEORY

Boundary Conditions

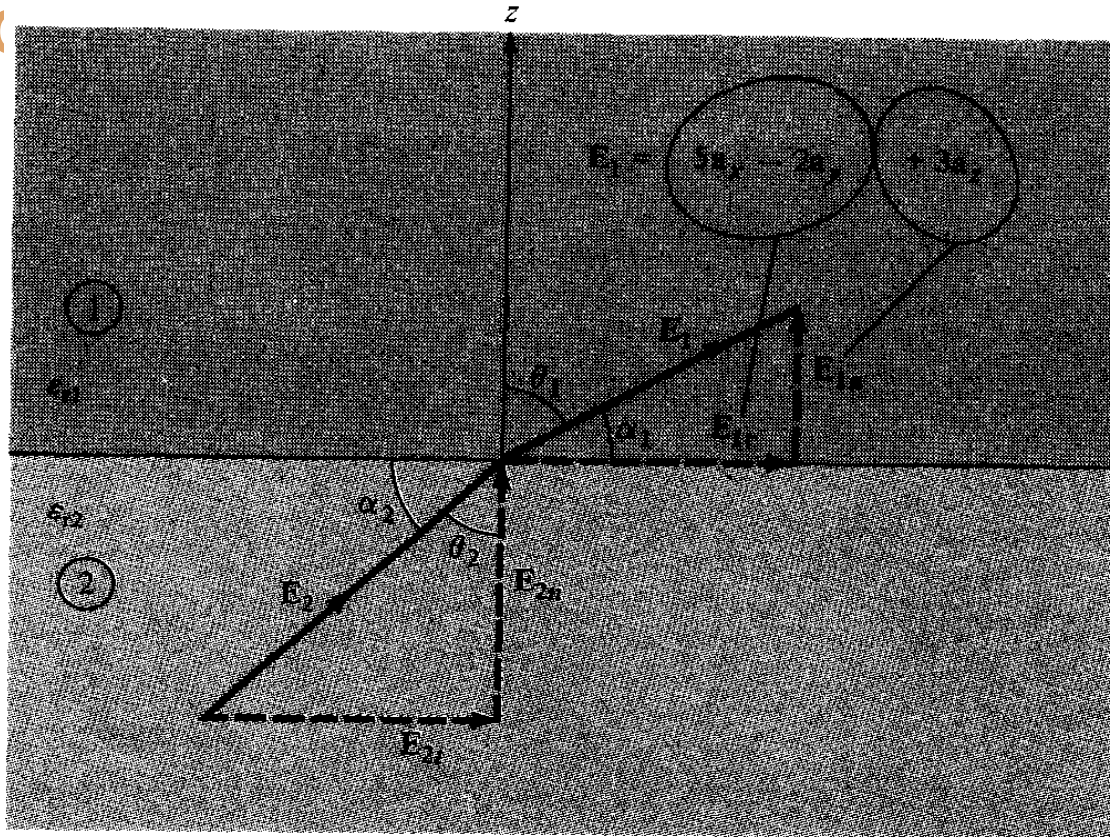
Cont d

Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\epsilon_{r1} = 4$ and for $z \leq 0$, $\epsilon_{r2} = 3$. A uniform electric field $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z$ kV/m exists for $z \geq 0$. Find

- (a) $\mathbf{E}_2$ for $z \leq 0$
- (b) The angles E_1 and E_2 make with the interface
- (c) The energy densities in J/m^3 in both dielectrics
- (d) The energy within a cube of side 2 m centered at $(3, 4, -5)$

ELECTRO MAGNETIC THEORY

Boundary Conditions



(a) Since $\mathbf{a}_z$ is normal to the boundary plane, we obtain the normal components as

$$E_{1n} = \mathbf{E}_1 \cdot \mathbf{a}_n = \mathbf{E}_1 \cdot \mathbf{a}_z = 3$$

$$\mathbf{E}_{1n} = 3\mathbf{a}_z$$

$$\mathbf{E}_{2n} = (\mathbf{E}_2 \cdot \mathbf{a}_z)\mathbf{a}_z$$

Also

$$\mathbf{E} = \mathbf{E}_n + \mathbf{E}_t$$

Hence,

$$\mathbf{E}_{1t} = \mathbf{E}_1 - \mathbf{E}_{1n} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

ELECTRO MAGNETIC THEORY

Boundary Conditions

Thus

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

Similarly,

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \rightarrow \epsilon_{r2}\mathbf{E}_{2n} = \epsilon_{r1}\mathbf{E}_{1n}$$

or

$$\mathbf{E}_{2n} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \mathbf{E}_{1n} = \frac{4}{3} (3\mathbf{a}_z) = 4\mathbf{a}_z$$

Thus

$$\begin{aligned}\mathbf{E}_2 &= \mathbf{E}_{2t} + \mathbf{E}_{2n} \\ &= 5\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z \text{ kV/m}\end{aligned}$$

ELECTRO MAGNETIC THEORY

Boundary Conditions

$$\alpha_1 = 90 - \theta_1$$

$$\alpha_2 = 90 - \theta_2$$

Since $E_{1n} = 3$ and $E_{1t} = \sqrt{25 + 4} = \sqrt{29}$

$$\tan \theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3} = 1.795 \rightarrow \theta_1 = 60.9^\circ$$

Hence,

$$\alpha_1 = 29.1^\circ$$

ELECTRO MAGNETIC THEORY

Boundary Conditions

Alternatively,

$$\mathbf{E}_1 \cdot \mathbf{a}_n = |\mathbf{E}_1| \cdot 1 \cdot \cos \theta_1$$

or

$$\cos \theta_1 = \frac{3}{\sqrt{38}} = 0.4867 \rightarrow \theta_1 = 60.9^\circ$$

Similarly,

$$E_{2n} = 4 \quad E_{2t} = E_{1t} = \sqrt{29}$$
$$\tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{29}}{4} = 1.346 \rightarrow \theta_2 = 53.4^\circ$$

Hence,

$$\alpha_2 = 36.6^\circ$$

ELECTRO MAGNETIC THEORY

Boundary Conditions

Note that $\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$ is satisfied.

(c) The energy densities are given by

$$\begin{aligned} w_{E1} &= \frac{1}{2} \epsilon_1 |\mathbf{E}_1|^2 = \frac{1}{2} \cdot 4 \cdot \frac{10^{-9}}{36\pi} \cdot (25 + 4 + 9) \times 10^6 \\ &= 672 \mu\text{J/m}^3 \end{aligned}$$

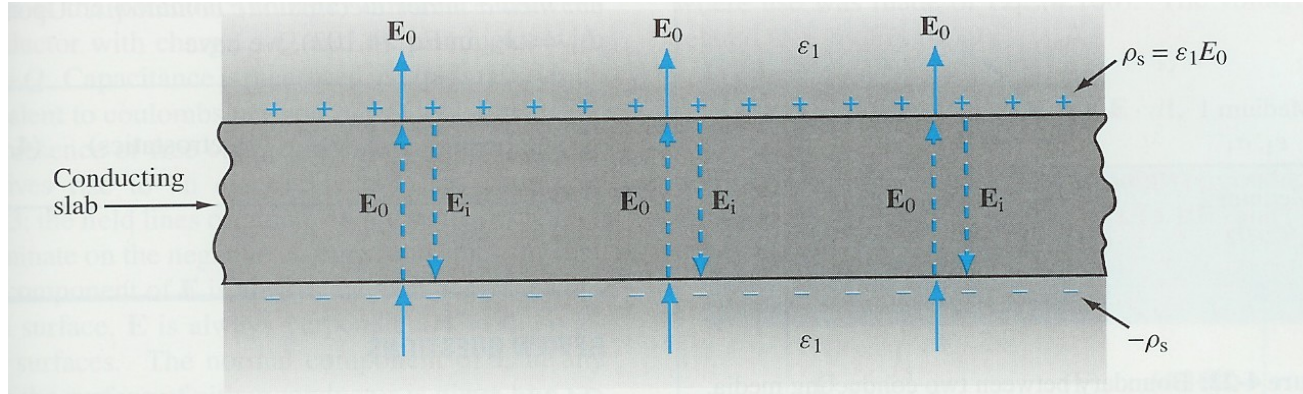
$$\begin{aligned} w_{E2} &= \frac{1}{2} \epsilon_2 |\mathbf{E}_2|^2 = \frac{1}{2} \cdot 3 \cdot \frac{10^{-9}}{36\pi} (25 + 4 + 16) \times 10^6 \\ &= 597 \mu\text{J/m}^3 \end{aligned}$$

(d) At the center (3, 4, -5) of the cube of side 2 m, $z = -5 < 0$; that is, the cube is in region 2 with $2 \leq x \leq 4$, $3 \leq y \leq 5$, $-6 \leq z \leq -4$. Hence

$$\begin{aligned} W_E &= \int w_{E2} dv = \int_{x=2}^4 \int_{y=3}^5 \int_{z=-6}^{-4} w_{E2} dz dy dx = w_{E2}(2)(2)(2) \\ &= 597 \times 8 \mu\text{J} = 4.776 \text{ mJ} \end{aligned}$$

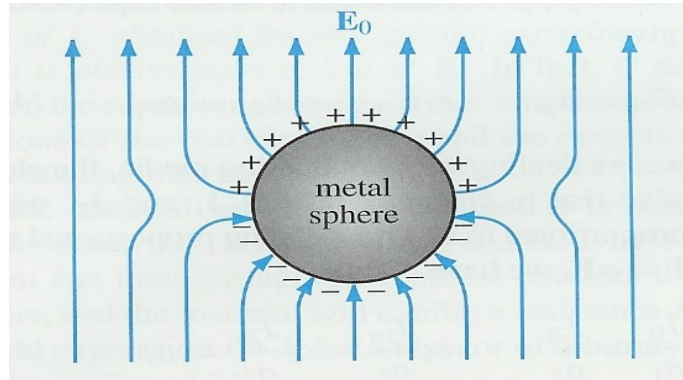
ELECTRO MAGNETI C THEORY

Di el ect r i c Conduct or Boundary



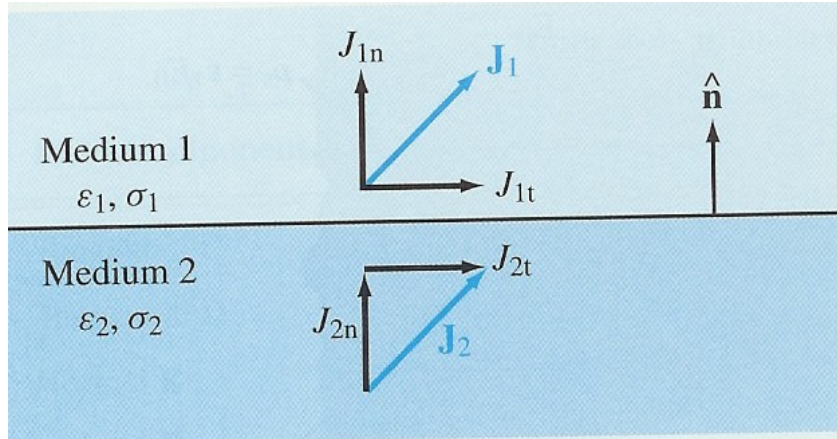
$$E_{1t} = E_{2t} = 0,$$

$$D_{1n} = \rho_s,$$



ELECTRO MAGNETIC THEORY

Conduct or Conduct or Boundary



$$\vec{J}_{1n} \left(\frac{\epsilon_1}{\sigma_1} - \frac{\epsilon_2}{\sigma_2} \right) = \rho_s,$$

$$E_{1t} = E_{2t}, \quad D_{1n} - D_{2n} = \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s,$$

$$\epsilon_1 \frac{J_{1n}}{\sigma_1} - \epsilon_2 \frac{J_{2n}}{\sigma_2} = \rho_s,$$

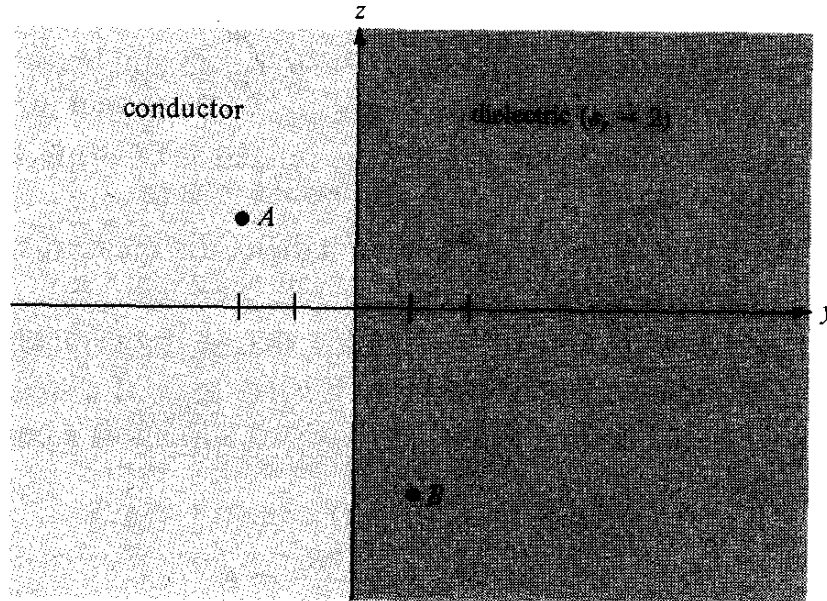
ELECTRO MAGNETIC THEORY

Conduct or Conduct or Boundary y

Region $y \leq 0$ consists of a perfect conductor while region $y \geq 0$ is a dielectric medium ($\epsilon_{1r} = 2$) as in Figure 5.16. If there is a surface charge of 2 nC/m^2 on the conductor, determine **E** and **D** at

(a) $A(3, -2, 2)$

(b) $B(-4, 1, 5)$



ELECTRO MAGNETIC THEORY

Conduct or Conduct or Boundary



(a) Point $A(3, -2, 2)$ is in the conductor since $y = -2 < 0$ at A . Hence,

$$\mathbf{E} = 0 = \mathbf{D}$$

(b) Point $B(-4, 1, 5)$ is in the dielectric medium since $y = 1 > 0$ at B .

$$D_n = \rho_s = 2 \text{ nC/m}^2$$

Hence,

$$\mathbf{D} = 2\mathbf{a}_y \text{ nC/m}^2$$

and

$$\begin{aligned}\mathbf{E} &= \frac{\mathbf{D}}{\epsilon_0 \epsilon_r} = 2 \times 10^{-9} \times \frac{36\pi}{2} \times 10^9 \mathbf{a}_y = 36\pi \mathbf{a}_y \\ &= 113.1 \mathbf{a}_y \text{ V/m}\end{aligned}$$

ELECTRO MAGNETIC THEORY

Assignment

A homogeneous dielectric ($\epsilon_r = 2.5$) fills region 1 ($x \leq 0$) while region 2 ($x \geq 0$) is free space.

(a) If $\mathbf{D}_1 = 12\mathbf{a}_x - 10\mathbf{a}_y + 4\mathbf{a}_z$ nC/m², find $\mathbf{D}_2$ and θ_2 .

(b) If $E_2 = 12$ V/m and $\theta_2 = 60^\circ$, find E_1 and θ_1 . Take θ_1 and θ_2 as defined in the previous example.

It is found that $\mathbf{E} = 60\mathbf{a}_x + 20\mathbf{a}_y - 30\mathbf{a}_z$ mV/m at a particular point on the interface between air and a conducting surface. Find $\mathbf{D}$ and ρ_s at that point.

ELECTRO MAGNETIC THEORY

Assignment Answers

(a) $12\mathbf{a}_x - 4\mathbf{a}_y + 1.6\mathbf{a}_z$ nC/m², 19.75°, (b) 10.67 V/m, 77°

$0.531\mathbf{a}_x + 0.177\mathbf{a}_y - 0.265\mathbf{a}_z$ pC/m², 0.619 pC/m².

ELECTRO MAGNETI C

THEORY Laplace and Poi son' s equat i on

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ELECTRO MAGNETI C THEORY

Lapl ace and Poi sson' s equat i on



To deri ve Lapl ace' s and Poi sson' s equat i ons , we start with Gauss' s Law in poi nt form :

$$\nabla \cdot \bar{D} = \nabla \cdot \epsilon \bar{E} = \rho_v$$

Use gradi ent concept :

$$\bar{E} = -\nabla V$$

ELECTRO MAGNETIC THEORY

Laplace and Poisson's equation

Cont d. ,

$$\nabla \cdot [\epsilon (-\nabla V)] = \rho_v$$

$$\nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon}$$

Oper at o

r :

$$\nabla \cdot \nabla = \nabla^2$$

ELECTRO MAGNETIC THEORY

Laplace and Poisson's equation

Cont d. ,

$$\nabla^2 V = -\frac{\rho_v}{\epsilon} \quad V/m^2$$

This is called
Poisson's
equation

applies to a
homogeneous
media.

When the free charge
density $\rho_v = 0$

$$\nabla^2 V = 0 \quad V/m^2$$

This is called
Laplace equation.

ELECTRO MAGNETIC THEORY

Laplace equation



Laplace's equation in Cartesian, cylindrical, or spherical coordinates respectively are given by

The above equation is Laplace equation in Rectangular coordinate system

ELECTRO MAGNETIC THEORY

Laplace equation Contd. ,



The above equation is Laplace equation in Cylindrical coordinate system

ELECTRO MAGNETIC THEORY

Laplace equation Cont d. ,



The above equation is
Laplace equation in
spherical coordinate system

ELECTRO MAGNETI C

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Solution to Laplace equation in one variable

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ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable



Example 1: Two infinite and parallel conducting planes are separated by a distance d , with one of the conductors in the $z = 0$ plane at $V = 0$ Volt and the other in the $z = d$ plane at $V = V_0$ Volt.

Assume

and
 ρ_s between the conductors.

Find : (a) V in the range $0 < z < d$

ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable Contd.,

a) Since $\rho_v = 0$ and the problem is in rectangular form, thus

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

We note that V will be a function of z only $V = V(z)$; thus

$$\begin{aligned}\nabla^2 V &= \frac{\partial^2 V}{\partial z^2} = 0 \\ \frac{d^2 V}{dz^2} &= \frac{d}{dz} \left(\frac{V}{dz} \right) = 0\end{aligned}$$

ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable Contd

Integrating twice :

$$\frac{dV}{dz} = A$$

$$V = Az + B$$

where A and B are constants and must be evaluated using given potential values at the boundaries :

$$V|_{z=0} = B = 0$$

$$V|_{z=d} = Ad = V_0$$
$$\rightarrow A = V_0 / d$$

ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable Contd.,

Substituting values of A and B we get,

$$\therefore V = \frac{V_0}{d} z \quad (V) \quad 0 < z < d$$

b)

$$\begin{aligned} \vec{E} &= -\nabla V = -\left(\hat{x} \frac{\partial V}{\partial x} + \hat{y} \frac{\partial V}{\partial y} + \hat{z} \frac{\partial V}{\partial z} \right) \\ &= -\hat{z} \frac{\partial V}{\partial z} = -\hat{z} \frac{V_0}{d} \quad (V / m) \end{aligned}$$

ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable Contd..

(c)

$$\bar{D} = \epsilon \bar{E} = -\hat{z} \frac{2\epsilon_0 V_0}{d} \quad (C / m^2)$$

ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable Contd...

(d) Surface charge :

$$\begin{aligned}\rho_s|_{z=0} &= \bar{D} \cdot \hat{n} = -\hat{z} \frac{2\varepsilon_0 V_0}{d} \cdot \hat{z} \\ &= -\frac{2\varepsilon_0 V_0}{d} \\ \rho_s|_{z=d} &= \bar{D} \cdot \hat{n} = -\hat{z} \frac{2\varepsilon_0 V_0}{d} \cdot (-\hat{z}) \\ &= +\frac{2\varepsilon_0 V_0}{d} \quad (C / m^2)\end{aligned}$$

ELECTRO MAGNETIC THEORY

Solution to Laplace equation in one variable Contd.,

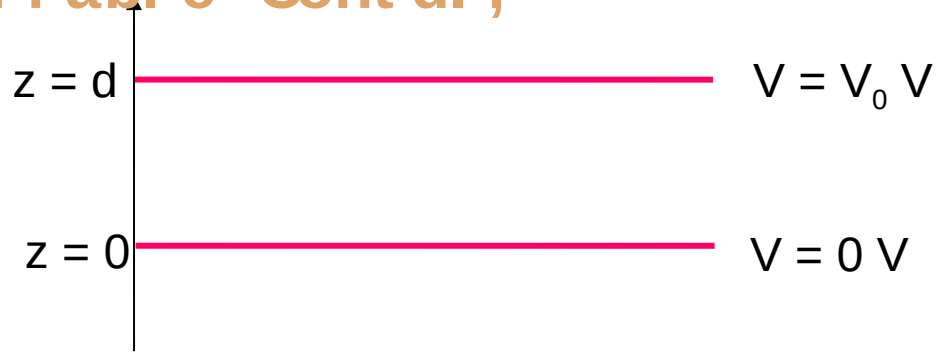
(e) Capacitance :

$$C = Q / V_{ab}$$
$$= \frac{|\rho_s ds|}{V_0}$$

$$\therefore C / m^2 = \frac{|\rho_s|}{V_0}$$
$$= \frac{2\varepsilon_0 V_0 / d}{V_0} = \frac{\varepsilon}{d} \quad (F / m^2)$$

ELECTRO MAGNETI C THEORY

Sol ut i on t o Lapl ace equat i on i n one
vari abl e Cont d. ,



ELECTRO MAGNETI C

THEORY

Solution to Poisson's equation

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ELECTRO MAGNETIC THEORY

Solution to Poisson's equation



Example 1: Two infinite and parallel conducting planes are separated by a distance d , with one of the conductors in the $x = 0$ plane at $V = 0$ Volt and the other in the $x = d$ plane at $V = V_0$ Volt.

Assume

and

between the conductors.

Find : (a) V in the range $0 < x < d$

ELECTRO MAGNETIC THEORY

Solution to Poisson's equation

Contd. ,

$V = f(x) :$

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}$$

$$\frac{d^2 V}{dx^2} = -\frac{\rho_0}{\epsilon}$$

$$\frac{dV}{dx} = -\frac{\rho_0}{\epsilon} x + A$$

$$V = -\frac{\rho_0}{\epsilon} \frac{x^2}{2} + Ax + B$$

ELECTRO MAGNETIC THEORY

Solution to Poisson's equation

Contd.,

Boundary condition :

$$V|_{x=0} = 0 = B$$

$$V|_{x=d} = V_0 = -\frac{\rho_0}{\epsilon} \frac{d^2}{2} + Ad$$

$$A = \frac{V_0}{d} + \frac{\rho_0}{\epsilon} \frac{d}{2}$$

$$V = -\frac{\rho_0}{\epsilon} \frac{x^2}{2} + Ax + B$$

ELECTRO MAGNETIC THEORY

Solution to Poisson's equation

Contd. ,

In region : $0 \leq x \leq d$

$$V = \frac{\rho_0}{\epsilon} \frac{x}{2} (d - x) + \frac{V_0}{d} x$$

$$\begin{aligned} \bar{E} &= - \frac{dV}{dx} \hat{x} \\ &= - \left[\frac{V_0}{d} + \frac{\rho_0}{\epsilon} \left(\frac{d}{2} - x \right) \right] \hat{x} \end{aligned}$$

ELECTRO MAGNETI C THEORY

Sol ut i on t o Poi sson' s equat i on

Cont d. ,
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Examp l e
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ELECTRO MAGNETIC THEORY

Solution to Poisson's equation

Contd.,
Example 2: Repeat the Previous
Example
with

Solution :

$$\nabla \cdot \vec{D} = \rho_v$$

$$\nabla \cdot \epsilon \vec{E} = 0$$

$$\frac{d}{dx} (1+x) \epsilon_0 \vec{E} = 0$$

$$\frac{d}{dx} (1+x) (-\nabla V) = 0$$

$$(1+x) \left(-\frac{dV}{dx} \right) = A$$

$$\left(-\frac{dV}{dx} \right) = \frac{A}{1+x}$$

$$V = -A \ln(1+x) + B$$

ELECTRO MAGNETIC THEORY

Solution to Poisson's equation

Cont'd

Boundary condition :

$$V|_{x=0} = 0 = B$$

$$V|_{x=d} = V_0 = -A \ln(1+d)$$

$$\rightarrow A = -\frac{V_0}{\ln(1+d)}$$

$$V = -A \ln(1+x) + B$$

ELECTRO MAGNETIC THEORY

Solution to Poisson equation Cont d. ,

In region : $0 \leq x \leq d$

$$V = V_0 \frac{\ln(1+x)}{\ln(1+d)}$$

$$\bar{E} = - \frac{dV}{dx} \hat{x} = - \frac{V_0}{(1+x)\ln(1+d)} \hat{x}$$

ELECTRO MAGNETI C

THEORY

Parallel Plate Capacitor and Cylindrical Coordinate Systems

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ELECTRO MAGNETIC THEORY

Parallel Plate Capacitor



Finding the capacitance of a parallel-plate capacitor

Steps:

- 1 Given V , use $E = -V$ to find E
- 2 Use $D = \epsilon E$ to find D
- 3 Evaluate D at either capacitor plate, $D = D_s =$
 D_n and
- 4 Recognize that $\epsilon_s = D \cdot n$
- 5 Find Q by a surface

$Q =$

Let Area of the plate is $S \text{ m}^2$

Now $Q =$

We know that $V = E \cdot d$

where E is the electric field

between the plates of a capacitor

separated by a distance d .

Now,

Hence,
$$C = \frac{|Q|}{V_0} = \frac{\epsilon \cdot S}{d}$$

Example: Two infinite length, concentric and conducting cylinders of radii a and b are located on the z axis. If the region between cylinders are charged free and

Solution : Use Laplace's equation in cylindrical coordinate. Find the capacitance per unit length.

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

ELECTRO MAGNETIC THEORY

Cylindrical Coordinates Contd. ,

$V = f(r)$ only :

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = 0$$

$$\frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = 0$$

$$\left(r \frac{\partial V}{\partial r} \right) = A$$

$$\frac{\partial V}{\partial r} = \frac{A}{r}$$

$$V = A \ln r + B$$

Boundary condition :

$$V|_{r=a} = V_0 = A \ln a + B$$

$$V|_{r=b} = 0 = A \ln b + B$$

Solving for A and B :

$$A = \frac{V_0}{\ln(a/b)} \quad ; \quad B = \frac{-V_0 \ln b}{\ln(a/b)}$$

Substituting the values for A and B We get ,

$$\therefore V = \frac{V_0 \ln(b/r)}{\ln(b/a)} \quad a < r < b$$

$$\bar{E} = -\nabla V = -\hat{r} \frac{\partial V}{\partial r} = \frac{V_0}{r \ln(b/a)} \hat{r}$$

$$\bar{D} = \epsilon \bar{E} = \frac{\epsilon V_0}{r \ln(b/a)} \hat{r}$$

Surface charge densities:

$$\rho_s|_{r=a} = \bar{D} \cdot \hat{r} = \frac{\epsilon V_0}{a \ln(b/a)}$$

$$\rho_s|_{r=b} = \bar{D} \cdot -\hat{r} = -\frac{\epsilon V_0}{b \ln(b/a)}$$

Line charge
densities :

$$\rho_{\ell}|_{r=a} = \rho_s|_{r=a} (2\pi a) = \frac{2\pi\epsilon V_0}{\ln(b/a)}$$

$$\rho_{\ell}|_{r=b} = \rho_s|_{r=b} (2\pi b) = - \frac{2\pi\epsilon V_0}{\ln(b/a)}$$

Capacitance per unit length:

$$C = \frac{Q}{V_{ab}} = \frac{|\rho_{\ell} d\ell|}{V_0}$$

$$C/m = \frac{|\rho_{\ell}|}{V_0} = \frac{2\pi\epsilon}{\ln(b/a)} \quad (F/m)$$

ELECTRO MAGNETI C

THEORY

Parallel Plate Capacitor and Spherical Coordinate Systems

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ELECTRO MAGNETIC THEORY

Parallel Plate Capacitor



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$Q =$

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Now $Q =$

We know that $V = E \cdot d$

where E is the electric field

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separated by a distance d .

Now,

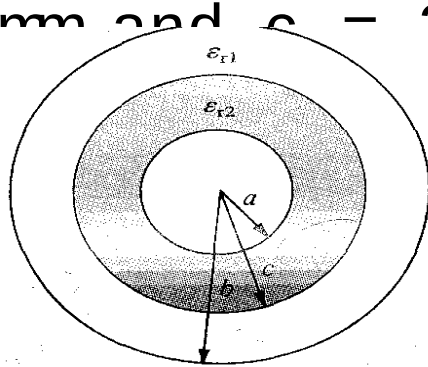
Hence,
$$C = \frac{|Q|}{V_0} = \frac{\epsilon \cdot S}{d}$$

ELECTRO MAGNETIC THEORY

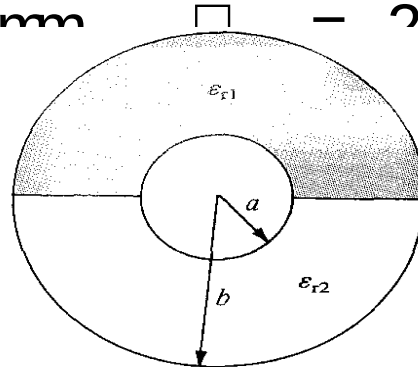
Spherical Coordinates

Example: The following figure represents cross section of 2 spherical capacitors, determine their capacitances. Let $a = 1\text{mm}$, $b = 3\text{mm}$ and $c = 2\text{mm}$ $\epsilon_1 = 2.5$ and $\epsilon_2 =$

$\epsilon_2 =$



(a)



(b)

ELECTRO MAGNETIC THEORY

Spherical Coordinate System

$$V = V_0 \cdot \frac{\frac{1}{r} - \frac{1}{b}}{\frac{1}{a} - \frac{1}{b}}$$

$$C = \frac{4 \pi \cdot \epsilon}{\frac{1}{a} - \frac{1}{b}}$$

ELECTRO MAGNETI C

THEORY

Cont i nui t y Equat i on

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ELECTRO MAGNETIC THEORY

Continuity Equation



The principle of conservation of charge:

“Charges can be neither created nor destroyed.”

But, equal amounts of positive and negative charge (pair of charges)

may be simultaneously created, obtained by separation, destroyed, or lost by recombination.

ELECTRO MAGNETIC THEORY

Continuity Equation Contd.,



$$I = \oint_S \mathbf{J} \cdot d\mathbf{S}$$

The above equation is known as Continuity Equation in Closed Surface

ELECTRO MAGNETIC THEORY

Integral Form of Continuity

Equation

Any outward flow of positive charge must be

balanced by a decrease of positive charge

(or perhaps an increase of negative charge) within the closed surface.

ELECTRO MAGNETIC THEORY

Integral Form of Continuity

Equation



If the charge inside the closed surface is denoted by Q_i ,

Then the rate of decrease is $-dQ_i/dt$ and

The total current is rate of change of charge.

ELECTRO MAGNETIC THEORY

Integral Form of Continuity Equation Contd. ,

$$I = \oiint_S \mathbf{J} \cdot d\mathbf{S} = - \frac{dQ_i}{dt}$$

The above equation is called Integral Form of the Continuity Equation.

ELECTRO MAGNETIC THEORY

Differential Form of Continuity

Equation

The differential form (or point form) of the continuity equation is obtained by

using the divergence theorem

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_{vol} (\nabla \cdot \mathbf{J}) dv$$

ELECTRO MAGNETIC THEORY

Differential Form of Continuity

Equation Contd. ,



We now represent Q_i by the volume integral of ρ_v :

$$\int_{vol} (\nabla \cdot \mathbf{J}) dv = - \frac{d}{dt} \int_{vol} \rho_v dv$$

ELECTRO MAGNETIC THEORY

Differential Form of Continuity Equation Contd. ,



If we keep the surface constant ,
the derivative becomes a partial
derivative. Writing it within the
integral ,

$$\int_{vol} (\nabla \cdot \mathbf{J}) dv = \int_{vol} - \frac{\partial \rho_v}{\partial t} dv$$

ELECTRO MAGNETIC THEORY

Differential Form of Continuity Equation Contd.,



$$(\nabla \cdot \mathbf{J})\Delta v = - \frac{\partial \rho_v}{\partial t} \Delta v$$

$$\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}$$

The above equation is called Differential Form (Point Form) of the Continuity Equation

ELECTRO MAGNETIC THEORY

Conductors and Dielectrics



If $\mathbf{J} = \frac{1}{r^3} (2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta)$ A/m², calculate the current passing through

- (a) A hemispherical shell of radius 20 cm
- (b) A spherical shell of radius 10 cm

ELECTRO MAGNETIC THEORY

Conductors Contd.,



$I = \int \mathbf{J} \cdot d\mathbf{S}$, where $d\mathbf{S} = r^2 \sin \theta d\phi d\theta \mathbf{a}_r$ in this case.

$$\begin{aligned} \text{(a) } I &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \frac{1}{r^3} 2 \cos \theta r^2 \sin \theta d\phi d\theta \bigg|_{r=0.2} \\ &= \frac{2}{r} 2\pi \int_{\theta=0}^{\pi/2} \sin \theta d(\sin \theta) \bigg|_{r=0.2} \\ &= \frac{4\pi}{0.2} \frac{\sin^2 \theta}{2} \bigg|_0^{\pi/2} = 10\pi = 31.4 \text{ A} \end{aligned}$$

(b) The only difference here is that we have $0 \leq \theta \leq \pi$ instead of $0 \leq \theta \leq \pi/2$ and $r = 0.1$. Hence,

$$I = \frac{4\pi}{0.1} \frac{\sin^2 \theta}{2} \bigg|_0^{\pi} = 0$$

Alternatively, for this case

$$I = \oint \mathbf{J} \cdot d\mathbf{S} = \int \nabla \cdot \mathbf{J} dv = 0$$

since $\nabla \cdot \mathbf{J} = 0$.

ELECTRO MAGNETI C

THEORY

Continuity Equation in differential form

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ELECTRO MAGNETIC THEORY

Integral Form of Continuity Equation Contd. ,



$$I = \oint_S \mathbf{J} \cdot d\mathbf{S} = - \frac{dQ_i}{dt}$$

The above equation is called Integral Form of the Continuity Equation.

ELECTRO MAGNETI C THEORY

Differential Form of Continuity

Equation

The differential form (or point form) of the continuity equation is obtained by

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ELECTRO MAGNETIC THEORY

Differential Form of Continuity

Equation Contd. ,



We now represent Q_i by the volume integral of ρ_v :

$$\int_{\text{vol}} (\nabla \cdot \mathbf{J}) dv = - \frac{d}{dt} \int_{\text{vol}} \rho_v dv$$

ELECTRO MAGNETIC THEORY

Differential Form of Continuity Equation Contd. ,



If we keep the surface constant ,
the derivative becomes a partial
derivative. Writing it within the
integral ,

$$\int_{vol} (\nabla \cdot \mathbf{J}) dv = \int_{vol} - \frac{\partial \rho_v}{\partial t} dv$$

ELECTRO MAGNETIC THEORY

Differential Form of Continuity Equation Contd.,



$$(\nabla \cdot \mathbf{J})\Delta v = - \frac{\partial \rho_v}{\partial t} \Delta v$$

$$\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}$$

The above equation is called Differential Form (Point Form) of the Continuity Equation

ELECTRO MAGNETI C

THEORY

Cont i nui t y Equat i on and Numer i cal Examp l e

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Depart ment of El ect r i cal and
El ect r on i cs Engi neer i ng

ELECTRO MAGNETIC THEORY

Integral Form of Continuity Equation

$$I = \oiint_S \mathbf{J} \cdot d\mathbf{S} = - \frac{dQ_i}{dt}$$

The above equation is called Integral Form of the Continuity Equation.

ELECTRO MAGNETIC THEORY

Differential Form of Continuity Equation Contd.,



$$\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}$$

The above equation is called
Differential Form (Point Form) of
the Continuity Equation

ELECTRO MAGNETIC THEORY

Numerical Example on Continuity Equation

The current density is given by $\mathbf{J} = \frac{1}{r} e^{-t} \mathbf{a}_r \text{ A/m}^2$.

- Total outward current at time instant $t = 1 \text{ s}$ and $r = 5 \text{ m}$.

$$I = J_r S_r = \left(\frac{1}{5} e^{-1} \mathbf{a}_r \right) (4\pi 5^2 \mathbf{a}_r) = \underline{\underline{23.11 \text{ A}}}$$

- Total outward current at time instant $t = 1 \text{ s}$ and $r = 6 \text{ m}$.

$$I = J_r S_r = \left(\frac{1}{6} e^{-1} \mathbf{a}_r \right) (4\pi 6^2 \mathbf{a}_r) = \underline{\underline{27.74 \text{ A}}}$$

ELECTRO MAGNETIC THEORY

Numerical Example on Continuity Equation Contd.,

- Finding volume charge density:

$$-\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r} e^{-t} \right) = \frac{1}{r^2} e^{-t}$$

$$\rho_v = - \int \frac{1}{r^2} e^{-t} dt = \frac{1}{r^2} e^{-t} + K(r)$$

$$t \rightarrow \infty, \rho_v \rightarrow 0 \Rightarrow K(r) = 0$$

$$\Rightarrow \rho_v = \frac{1}{r^2} e^{-t} \text{ C/m}^3 \Rightarrow v_r = \frac{J_r}{\rho_v} = \underline{\underline{r \text{ m/s}}}$$

ELECTRO MAGNETI C THEORY

Rel axat i on Ti me

$$\mathbf{J} = \sigma \mathbf{E}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}$$

$$\nabla \cdot \sigma \mathbf{E} = \frac{\sigma \rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$$

ELECTRO MAGNETIC THEORY

Relaxation Time Cont d. ,



$$\nabla \cdot \sigma \mathbf{E} = \frac{\sigma \rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$$

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon} \rho_v = 0$$

$$\frac{\partial \rho_v}{\rho_v} = -\frac{\sigma}{\epsilon} \partial t$$

ELECTRO MAGNETIC THEORY

Relaxation Time Cont d. ,

and integrating both sides gives

$$\ln \rho_v = -\frac{\sigma t}{\epsilon} + \ln \rho_{v0}$$

where $\ln \rho_{v0}$ is a constant of integration. Thus

$$\rho_v = \rho_{v0} e^{-t/T_r}$$

$$T_r = \frac{\epsilon}{\sigma}$$

ELECTRO MAGNETIC THEORY

Relaxation Time Cont d. ,

ρ_{v0} is the initial charge density (i.e., ρ_v at $t = 0$).

Relaxation time is the time it takes a charge placed in the interior of a material to drop to $e^{-1} = 36.8$ percent of its initial value.

It is short for good conductors and long for good dielectrics.

ELECTRO MAGNETIC THEORY

Relaxation Time Contd. ,

For Copper with $\sigma = 5.8 \times 10^7$ mhos/m Find relaxation time when $\epsilon_r = 1$.

$$T_r = \frac{\epsilon_r \epsilon_0}{\sigma} = 1 \cdot \frac{10^{-9}}{36\pi} \cdot \frac{1}{5.8 \times 10^7}$$

$$= 1.53 \times 10^{-19} \text{ s}$$

ELECTRO MAGNETIC THEORY

Relaxation Time Cont d. ,

For Quartz with $\sigma = 10^{-17}$ mhos/m
Find relaxation time when $\epsilon_r = 5$.

$$\begin{aligned} T_r &= 5 \cdot \frac{10^{-9}}{36\pi} \cdot \frac{1}{10^{-17}} \\ &= 51.2 \text{ days} \end{aligned}$$

ELECTRO MAGNETIC THEORY

Assignment



For Copper with $\sigma = 5.8 \times 10^7$ mhos/m Find relaxation time when $\tau = 1$.

For Quartz with $\sigma = 10^{-17}$ mhos/m Find relaxation time when $\tau = 5$.

For the current density $\mathbf{J} = 10z \sin^2 \phi \mathbf{a}_\rho$ A/m², find the current through the cylindrical surface $\rho = 2$, $1 \leq z \leq 5$ m.



THANK YOU

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